

REVIEW ARTICLE

A Survey on Deep Learning for Cellular Traffic Prediction

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With the widespread deployment of 5G networks and the proliferation of mobile devices, mobile network operators are confronted not only with massive data growth in mobile traffic, but also with highly complex and dynamic traffic patterns. Given these challenges to network operation, cellular traffic prediction is becoming an essential network capability for ensuring quality of service and reducing costs. Accurate and timely cellular traffic prediction is essential for resource allocation, base station energy conservation, and network optimization. Recent years have seen widespread adoption of deep-learning-based models for cellular traffic prediction, with notable performance improvements. This survey encompasses representative data, model architectures, and state-of-the-art performance to provide a comprehensive account of deep learning techniques for cellular traffic prediction. After defining the problem of cellular traffic prediction and describing the available data, we describe in detail how deep learning techniques are used to capture the most crucial temporal and spatial dependencies of cellular traffic. We then summarize the state-of-the-art performance on 2 popular open datasets with multiple data settings to facilitate the comparison of deep-learning-based methods. Finally, we briefly outline the applications of cellular traffic prediction and discuss the remaining challenges and future research directions.

Introduction

Internet-connected mobile devices impact many aspects of an individual's life and work, including social contact, commerce, and travel. The proliferation of smartphones has led to a surge in cellular traffic data, creating a huge demand for network capacity and presenting management and scheduling challenges for cellular network operation. As a crucial aspect of telecom network operation, traffic prediction plays an important role in resource allocation, network planning, and network optimization [1]. Specifically, accurate traffic forecasting enables efficient dynamic resource allocation in cellular networks, thus decreasing packet loss rates and network congestion. Traffic prediction for base stations can facilitate the automatic switching of sleep and wake modes and thus ensure energy savings without compromising the user experience [2–4].

Cellular traffic prediction problems can be classified into temporal and spatial-temporal prediction problems. The temporal prediction problem considers only the inherent temporal dependencies of the traffic data. The spatial-temporal prediction problem considers not only temporal correlations in the traffic data but also spatial correlations between the traffic data of multiple network elements. In this survey, both these problems are considered. Although the challenges to cellular traffic prediction have been extensively studied for many years, they remain unresolved mainly due to the following reasons:

Temporal dependencies: Cellular traffic data are generally represented as a predictable time series. Commonly observed periodicity, trend, and other characteristics of time series are indications of normal temporal correlations. The past values of

a traffic data sequence influence its future values. Moreover, the traffic data values of different future moments are differentially influenced by historical observations at various times, resulting in dynamic temporal dependencies. In addition, cellular traffic data are extremely dynamic compared to road traffic due to the unpredictability of user behavior [5]. For example, a user can be utilizing a video app that consumes a large amount of data at one moment, and then suddenly disconnect due to an unstable network signal or the user's own behavior. Then, the corresponding traffic data vanishes immediately.

Spatial dependencies: Due to their own work and living needs, users may actively travel to other locations in the city, such as dwellings, companies, subways, and hospitals, as shown in Fig. 1, resulting in traffic transfer between multiple cell towers. This results in the formation of the spatial correlations between base stations in the cellular network. In addition, to improve user experience, the cellular network performs passive user handover scheduling, for example, by switching customers to a distant cell tower with a stronger signal or a close cell tower with fewer users. The unpredictability of user behavior and the complexity of the scheduling mechanism of the cellular network introduce complicated spatial correlations between cell towers [6]. Consequently, cellular traffic data exhibit both local and global spatial dependencies. Moreover, it may incorporate dynamic spatial correlations that change over time, as shown in Fig. 1.

This survey focuses on a data-driven method for cellular traffic prediction, which is essentially time-series forecasting. Traditional time-series forecasting algorithms, such as autoregressive integrated moving average (ARIMA) [7], have difficulty

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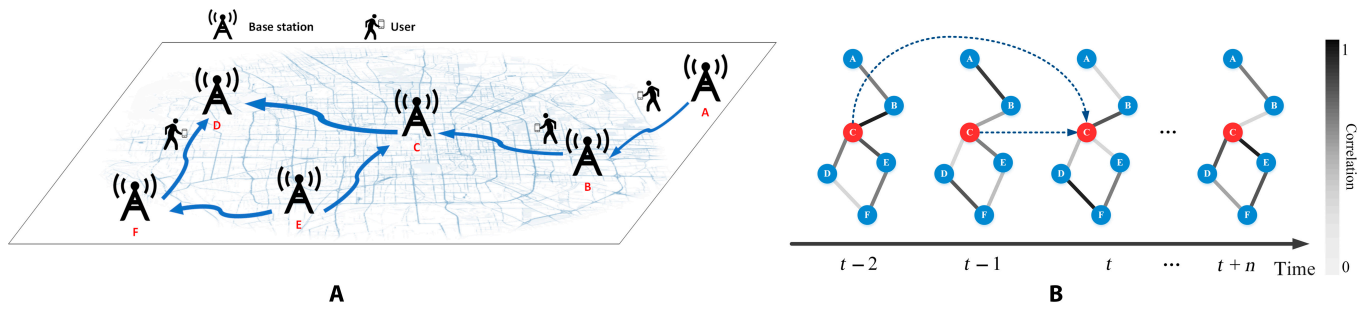


Fig. 1. Spatial-temporal correlations in cellular traffic network. (A) Schematic illustration of a cellular network with users switching between different base stations. (B) Traffic data of the central base station C is influenced not only by its own historical data but also by the traffic data of its spatial neighbors; blue dotted lines represent temporal correlations, solid lines denote spatial correlations, and the bar on the right represents the strength of the spatial correlation.

capturing the nonlinear temporal correlations of traffic data. Researchers have proposed the use of machine learning algorithms, such as Gaussian process regression (GPR) [8], support vector regression [9], and tree-based LightGBM [10], to address these challenges. The overall prediction performance of machine learning algorithms is preferable to that of statistical models [11]. Specifically, according to the findings in [11], machine learning methods exhibit better prediction performance and scalability compared to statistical approaches. Moreover, statistical methods face challenges when applied to large-scale time-series prediction problems. However, it is crucial to note that machine learning methods require more data and computational resources. Moreover, machine learning algorithms rely heavily on hand-crafted features and have difficulty capturing the complicated data patterns of cellular traffic, which are influenced by spatial and temporal correlations. Recently, deep learning [12] has developed considerably, demonstrating remarkable performance across a range of tasks [13–15]. Researchers have shifted their interest to implementing deep learning for cellular traffic prediction. Alawe et al. [16] investigated using a long short-term memory (LSTM) neural network [17] for cellular traffic prediction. Wang et al. [18] employed an LSTM network to learn long-term dependencies in cellular traffic and utilized GPR to predict the residual random components, thereby producing a relatively long-term and accurate traffic prediction. Deep learning has the potential to learn the intricate patterns of traffic data due to its powerful representation learning capability.

However, the aforementioned approaches to sequence modeling neglect spatial correlations, resulting in loss of accuracy. Several studies [3,19] have proposed the use of convolutional LSTM (ConvLSTM) [20], which integrates a convolutional neural network (CNN) [21] with LSTM, to address the spatial-temporal traffic prediction problem. Specifically, the methodologies based on ConvLSTM employ a grid-based approach and assume that there is a Euclidean spatial relationship between network elements. Nevertheless, this strategy has limitations because the spatial relationship between cellular network elements is more graph-based than grid-based. With the introduction of the graph neural network (GNN) [22], deep learning has begun to demonstrate considerable promise in graph modeling. The GNN has demonstrated enormous potential in graph node classification [23], recommendation systems [24], and road traffic prediction [25] and has substantially improved the performance of applications in multiple fields. Recently, many academics have shifted their focus to GNN-

based cellular traffic prediction and obtained promising results [26–29].

Existing surveys have outlined the application of deep learning to mobile networks [30,31]. An encyclopedic review of mobile and wireless networking research based on deep learning was presented in [30]. Jiang [11] reviewed machine-learning-based cellular traffic prediction and discussed statistical, machine learning, and deep learning models. However, that review does not analyze the most influential factors of traffic prediction or the associated deep learning techniques, nor does it investigate the growing use of GNNs in this field. None of the aforementioned surveys presents a systematic review of the key technologies of deep learning implemented in cellular traffic prediction. Therefore, this paper aims to do so. Its purpose is to serve as a guide for interested readers, especially those who wish to explore the application of deep learning techniques to unsolved problems in cellular traffic prediction.

The main contributions of this survey are summarized as follows:

- We provide a categorization of cellular traffic prediction problems and list some available datasets to assist readers in comprehending the surveyed issues and traffic data.
- We present a taxonomy for existing deep-learning-based temporal and spatial-temporal traffic prediction approaches, outlining their key design choices, to enable academics to comprehend the status quo and inspire the development of more innovative models.
- We summarize the state-of-the-art performance on popular open datasets with multiple data settings and list some open-source codes to facilitate the comparison of deep learning-based solutions.

- We present a summary of the applications of cellular traffic prediction and discuss challenges and future research directions to guide researchers interested in exploring this frontier field.

The remainder of this survey is organized as follows. In the “Related Surveys” section, we introduce and compare our work with related surveys. We define the problem of cellular traffic prediction and describe open traffic datasets in the “Problem Definition and Datasets” section. In the “Deep Learning Methods” section, we discuss the deep learning techniques used for traffic prediction and summarize the state-of-the-art performance, as well as provide a comprehensive list of open-source codes. In the “Applications” section, we outline the applications of cellular traffic prediction. In the “Challenges and Future Directions” section, we provide guidance for follow-up studies. Finally, we end with “Conclusions”.

Related Surveys

An abundance of surveys and guides on the subject of wireless networks have appeared in recent years. Different aspects of the logic of deep learning approaches for assessing dynamic wireless networks were investigated in [31]. The authors explored many deep learning applications at multiple layers, and investigated the improvement of hardware resource allocation, network security, and related topics. They also presented 10 difficulties involving wireless networking based on deep learning to stimulate the curiosity of future researchers. Zhang et al. [30] provided a comprehensive overview of mobile network analysis using deep learning techniques. On the basis of deep learning, they provided a detailed review of related research on mobility analysis, user localization, network control, network security, etc. Kulin et al. [32] examined existing research on enhancing network performance with machine learning techniques. They discussed the physical, medium access control, and network layers, describing how machine learning techniques can be implemented in a variety of network applications. For other relevant surveys and debates, please refer to [33,34] and the references therein. In the field of cellular traffic prediction, Jiang [11] summarized the key research and classified the prediction challenges as temporal and spatial-temporal modeling problems. Nonetheless, that review focused mainly on the presentation of the principles of the methods rather than their application in the field of cellular traffic prediction. In particular, although graph-based models are an important methodology for modeling spatial-temporal cellular traffic prediction, they are only discussed within a limited scope.

As far as we know, despite the fact that important attempts have been made in the field of cellular traffic prediction, few studies have comprehensively reviewed the evolution and current conditions of mobile network traffic prediction. This survey aims to fill the gap between deep learning methodologies and cellular traffic prediction, mainly focusing on deep-learning-based methodologies, problems, challenges, and future directions.

Problem Definition and Datasets

In this section, we provide a classification of cellular traffic prediction problems, as well as a general problem formulation to aid the reader's understanding of the prediction task. Some publicly available traffic datasets are also introduced and described accordingly. We aim to provide the preliminaries for the deep-learning-based algorithms introduced in subsequent sections.

Problem definition

Cellular traffic is typically collected at regular intervals, representing the time-varying network status of cellular networks. Therefore, predicting cellular traffic is fundamentally a time-series prediction problem. In this paper, we focus on data-driven approaches that predict traffic values in future time steps using historical data. We can also formulate this problem as a supervised learning problem by applying moving windows on the time-series data to produce multiple input-output pairs as training data. In this manner, it is possible to predict cellular traffic by learning a model that maps historical traffic data with fixed time slots to future traffic with defined time steps. Then, methods based on artificial intelligence,

including deep learning, can be used to learn the complicated mapping relationships.

Cellular traffic prediction problems can be classified into 2 main types: temporal and spatial-temporal. This paper presents the temporal prediction problem as a univariate temporal prediction problem, such as predicting the traffic flow of a single base station based solely on its own historical traffic data, which only considers temporal dependencies. The spatial-temporal prediction problem takes into account both spatial and temporal dependencies in traffic data and aims to predict the traffic data series of multiple network elements with spatial dependencies. For example, predicting the traffic flow of multiple base stations in a cellular network, which often influence each other spatially due to user hand-over, is a spatial-temporal prediction problem.

Multiple types of indicators, including but not limited to upstream and downstream traffic flow [35], the number of connected users [36], and the number of radio resource control connections [37], are frequently applied to monitor the states of cellular networks. Based on the number of forecasting target variables, cellular traffic prediction problems can alternatively be divided into univariate and multivariate prediction problems. The objective of univariate prediction is to forecast a single variable. There have been multivariate prediction attempts to simultaneously forecast several network indicators (variables), and there are frequently serial correlations between the multiple predictive variables. Predicting the number of connected users and traffic flow of a base station concurrently is a typical multivariate prediction problem, as an increase in the number of connected users generally causes a rise in traffic flow.

From the perspective of prediction length, traffic prediction can also be divided into 2 scales: short-term and medium-to-long term. This classification lacks a single criterion due to the influence of time granularity and other factors. In general, when predicting a time series with a 5-min granularity, it can be classified based on the prediction length as short-term (5 to 30 min) or medium-to-long term (over 30 min) [25]. Regarding traffic prediction time steps, cellular traffic prediction problems can be classified as either single- or multi-step prediction problems, corresponding to predicting only the traffic in the next time step or over multiple future time steps.

Despite different mathematical notations and representations, the problem formulations of the majority of existing cellular traffic forecasting works are comparable. We provide a general formulation for the temporal prediction problem by summarizing such works. Given the traffic data of T historical time steps $(\mathbf{x}_{t-T+1}, \mathbf{x}_{t-T}, \dots, \mathbf{x}_t)^T \in \mathbb{R}^{T \times F}$, where F denotes the feature dimension and t indicates the current time step, $\mathbf{x}_t$ represents the traffic observations at time step t . The target is to predict the traffic data in the next τ time steps $(\hat{\mathbf{x}}_{t+1}, \hat{\mathbf{x}}_{t+2}, \dots, \hat{\mathbf{x}}_{t+\tau})^T \in \mathbb{R}^{\tau \times F}$. We formulate the temporal prediction problem as learning a mapping function $\mathcal{F}$ to map historical traffic data with fixed time steps to future traffic with defined time steps:

$$(\hat{\mathbf{x}}_{t+1}, \hat{\mathbf{x}}_{t+2}, \dots, \hat{\mathbf{x}}_{t+\tau}) = \mathcal{F}_{\theta}(\mathbf{x}_{t-T+1}, \mathbf{x}_{t-T}, \dots, \mathbf{x}_t), \quad (1)$$

where θ denotes the learnable parameters in the prediction model.

For the formulation of the spatial-temporal prediction problem, we first define a directed or undirected graph $\mathcal{G} = (V, E, A)$ to represent the spatial correlations, as they are typically represented by graphs in the cellular traffic prediction problem.

V represents the set of network elements that produce traffic data, where $|V| = N$ (N indicates the number of elements). We may also call them nodes in this paper. E is the set of edges and denotes the spatial correlations between nodes. $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix of the graph $\mathcal{G}$. The traffic indicator observations defined on graph $\mathcal{G}$ are denoted as $\mathbf{X}_t = (\mathbf{x}_t^1, \mathbf{x}_t^2, \dots, \mathbf{x}_t^N)^T \in \mathbb{R}^{N \times F}$, where $\mathbf{x}_t^i$ denotes the observations of the i th network element at time step t . Given the graph traffic matrices of T historical time steps $\mathcal{X} = (\mathbf{X}_{t-T+1}, \mathbf{X}_{t-T}, \dots, \mathbf{X}_t) \in \mathbb{R}^{T \times N \times F}$, we formulate the spatial-temporal prediction problem as learning a mapping function F to predict the future graph traffic matrices in the subsequent τ time steps $\mathcal{Y} = (\hat{\mathbf{X}}_{t+1}, \hat{\mathbf{X}}_{t+2}, \dots, \hat{\mathbf{X}}_{t+\tau}) \in \mathbb{R}^{\tau \times N \times F}$:

$$(\hat{\mathbf{X}}_{t+1}, \hat{\mathbf{X}}_{t+2}, \dots, \hat{\mathbf{X}}_{t+\tau}) = F_{\theta}(\mathbf{X}_{t-T+1}, \mathbf{X}_{t-T}, \dots, \mathbf{X}_t; \mathcal{G}), \quad (2)$$

where θ still denotes the learnable parameters in the model. Additionally, we have $F = 1$ for the univariate prediction problem and $F > 1$ for the multivariate prediction problem. Likewise, for the multi-step prediction problem, we have $\tau > 1$.

Datasets

This section focuses on describing widely used public datasets in the field of cellular traffic prediction. Open datasets provide a suitable test bed for methods and techniques designed to address the cellular traffic prediction problem.

Milan [35]: This dataset provides an unprecedented amount of real-world telecommunication data from the “Telecom Italia Big Data Challenge”. The data were gathered in Milan and the geographical area was split into 100×100 regions. Each region has a dimension of 235×235 m and is referred to as a cell. The dataset is composed of 5 types of call detail records (CDRs), namely, received SMS, sent SMS, incoming call, outgoing call, and internet. The Milan dataset [35] contains cellular traffic data for the period from 2013 November 1 to 2014 January 1 with a time interval of 10 min. A raw version of the Milan dataset is available at: <https://doi.org/10.7910/DVN/EGZHFV>. Zhang et al. [38] provide an open-source processed version of the Milan dataset. They integrated the original 10-min scale to an hourly scale, aggregated the traffic of SMS-in and SMS-out into SMS, and aggregated the traffic of call-in and call-out into call.

Trentino [35]: This dataset was collected in Trentino and was also made available by the “Telecom Italia Big Data Challenge”. The geographical area of Trentino was divided into a grid of 117×98 regions. This dataset, like the Milan dataset, provides 5 types of CDR data gathered between 2013 November 1 and 2014 January 1, with a 10-min data granularity. A raw version of the Trentino dataset is available at: <https://doi.org/10.7910/DVN/EGZHFV>.

Other public cellular traffic datasets are only infrequently utilized for study, so we list them without extensive explanation: Hsinchu [39], C2TM [40], and Shanghai [41].

Deep Learning Methods

In this section, we conduct a systematic review of deep-learning-based solutions for temporal and spatial-temporal cellular traffic prediction. We introduce the principles, advantages, and limitations of the relevant deep learning techniques employed for traffic prediction. As shown in Fig. 2, we summarize a

taxonomy of key deep learning techniques for cellular traffic prediction. We also summarize and classify deep-learning-based models for spatial and temporal dependency modeling for cellular traffic prediction, as well as the relevant studies, as shown in Table 1. The state-of-the-art performance of traffic prediction methods on the 2 most widely used public datasets with multiple data settings is presented and discussed. A list of open-source codes is presented to facilitate the comparison of different solutions. This section aims to give readers a systematic understanding of deep-learning-based traffic prediction methods.

Temporal prediction

This section focuses on the complicated temporal dependencies of cellular traffic that have been studied in the literature. Temporal dependencies are the complex and nonlinear interactions that exist between traffic measurements at different time steps. The temporal dependencies are influenced not only by the nearby time slots but also by longer time periods. We examine and classify the various types of temporal dependency modeling approaches obtained in our survey for cellular traffic data prediction, which include recurrent neural network (RNN)-, CNN-, and attention-based methods.

RNN-based methods

RNNs are a popular method for handling time-series issues. As a result, they are frequently used to predict cellular traffic to capture temporal dependencies. RNNs are a particular class of neural network that have hidden states and allow the use of previous outputs as inputs. Their capacity to capture long-term dependencies of sequential data is, however, constrained by the gradient vanishing/exploding problem brought on by the architecture of the original RNN. The LSTM network together with gated recurrent unit (GRU) was developed to address this issue, and it quickly became one of the most-used deep learning prediction techniques for mobile traffic data.

LSTM is a modified RNN designed to overcome the problem of long-range dependencies that RNNs cannot manage. It blends short-term and long-term memory through nuanced gate control, thereby partially resolving the gradient vanishing/exploding issue. An LSTM unit has 3 gates: a forget gate, an input gate, and an output gate. At time t , it receives the observation vector $\mathbf{x}_t$ and a tuple $(\mathbf{c}_{t-1}, \mathbf{h}_{t-1})$, where $\mathbf{c}_{t-1}$ represents the cell state at time $t-1$, and $\mathbf{h}_{t-1}$ represents the hidden state. The outputs of these 3 gates are as follows:

$$\begin{aligned} \mathbf{f}_t &= \sigma(\mathbf{W}_f \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \\ \mathbf{i}_t &= \sigma(\mathbf{W}_i \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i), \\ \tilde{\mathbf{C}}_t &= \tanh(\mathbf{W}_C \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_C), \\ \mathbf{C}_t &= \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{C}}_t, \\ \mathbf{o}_t &= \sigma(\mathbf{W}_o \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \\ \mathbf{h}_t &= \mathbf{o}_t \odot \tanh(\mathbf{C}_t), \end{aligned} \quad (3)$$

where $\mathbf{W}$ and $\mathbf{b}$ denote the weight matrix and bias, respectively; $\tilde{\mathbf{C}}_{t-1}$ represents the candidate of the cell state $\mathbf{C}_t$; σ denotes the sigmoid function; $\mathbf{f}_t$, $\mathbf{i}_t$, and $\mathbf{o}_t$ denote the forget, input, and output gates, respectively; and $\odot$ denotes the sigmoid activation function. A gate permits information to pass selectively using the sigmoid function as a selection tool and the tanh function

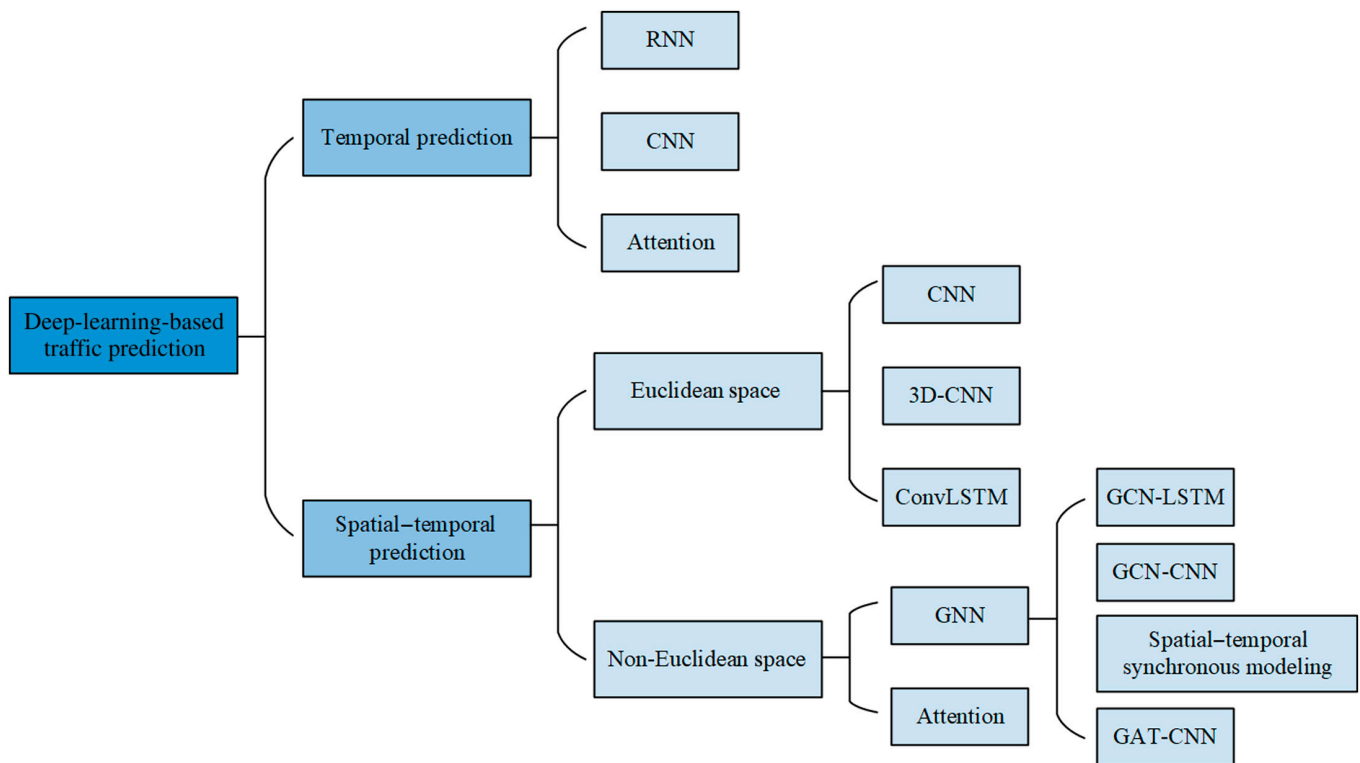


Fig. 2. Taxonomy of key deep learning techniques for cellular traffic prediction.

as a transformation tool. These 2 functions together implement the 3 gates.

Due to their shared weights and recurrent architecture, RNNs have been demonstrated to generate relatively accurate short-term traffic predictions [57]. However, as discussed above, RNNs are not ideal for modeling long-term temporal dependencies. In light of this, LSTM and variants of LSTM are offered, including gated units and cell variables that are carefully built to store past data. In this regard, a large amount of research has been conducted. Zhu and Wang [2] used LSTM to forecast the distribution of cellular traffic in a certain service region, enabling intelligent base station sleeping and lowering energy consumption. A hybrid model comprising double seasonal ARIMA (D-SARIMA) and LSTM was developed by [51]. D-SARIMA was presented to capture the double seasonality of cellular traffic data, whereas LSTM models the nonlinear residual components of the data. Simulations demonstrate that the blended model outperforms both ARIMA and LSTM, in part because it concurrently addresses seasonality and nonlinearity. Fawaz and Mosleh [46] estimated the volume of cellular traffic using LSTM. Their proposed model incorporates min-max normalization and single-exponential smoothing to handle bursts and large fluctuations. Wang et al. [53] proposed a data augmentation model for cellular traffic generation, together with an LSTM-based model for cellular traffic prediction. Wang et al. [18] suggested integrating LSTM and GPR methodologies to attain precise cellular traffic prediction at the single-cell level. In [52], an LSTM network was combined with discrete Fourier transform (DFT) to model the most relevant features inherited in the time series. In general, LSTM has been widely employed for cellular traffic forecasting and has demonstrated superior performance compared

to other approaches. For more current investigations of LSTM sequence modeling, see, for instance, [43–45,50,54].

CNN-based methods

Despite their widespread use, the multiple gate mechanisms of LSTM may introduce considerable temporal complexity and decrease the performance of the entire model. The CNN [58], which was initially proposed for image processing, has been used as a non-sequential modeling technique for cellular traffic forecasting, and has the advantages of simpler structure and faster operation over RNN-based methods.

Typically, CNN-based approaches use separate channels to model discrete time periods, such as daily and weekly periods. When dealing with sequences of cellular traffic, it is worth noting that 1D convolution has also been widely used. By using a CNN for training, the model is able to thoroughly examine the traffic patterns on the time scale, thereby enhancing its ability to predict cellular traffic data.

Gao et al. [37] proposed a CNN-based model with 2 channels of traffic input and a residual network (ResNet) structure, each constructed to capture temporal patterns in cellular traffic. ResNet employs residual learning and creates shortcut connections between the input and output to permit deeper neural networks while preserving the precision and processing efficiency of the model (see also [59] for similar research).

An important variation is the temporal convolutional network (TCN) [60], which is based on the CNN and uses both causal and dilated convolution. First, to guarantee that a sequence of any length can be input and mapped to a sequence of the same length, the TCN creates a 1D fully convolutional network (FCN) [61] with zero padding and a kernel size of 1×1 . Second, the TCN employs casual convolution to ensure

Table 1. Temporal and spatial–temporal modeling methods for traffic prediction

Category	Method	Model	Relevant studies
Temporal prediction	RNN-based	RNN	[57]
		LSTM	[2,16,42–50]
		LSTM+D-SARIMA	[51]
		LSTM+FFT	[52]
		LSTM+GAN	[53]
		LSTM+GPR	[18]
		MuLSTM	[54]
		ResNet	[37]
	CNN-based	TCN	[39]
		Attention only	[62]
	Attention-based	CNN+Attention	[55]
		RNN+Attention	[56]
Spatial–temporal prediction	Grid-based	2D-CNN	[38,68,93]
		3D-CNN	[63,64,66]
		ConvLSTM	[3,19,56,66,67,75,79,82]
	Graph-based	GCN+LSTM	[4,76]
		GCN+GRU	[29]
		GCN+1D CNN	[26]
		GCN+GLU	[27]
		GAT+TCN	[39]
	Spatial–temporal synchronous modeling	STConv+Attention	[28]
		Aggregation GCN	[92]
	Attention-based	Structured RNN+Attention	[77]
		GCN+CNN+Attention	[78]
		CNN+RNN+Attention	[80]
		ConvLSTM+Attention	[79]

that the prediction output y_t depends exclusively on the previously observed inputs $x_0, \dots, x_{t-1}$ and that there is no “leakage” of information from the future to the past. Consequently, the TCN may be expressed simply as TCN = 1D FCN + casual convolutions [60]. Then, Bai et al. [60] adopted dilated convolutions to provide an exponentially expanding receptive field to satisfy the needs of a lengthy historical sequence. The definition of dilated convolution is

$$F(s) = (\mathbf{x} *_{d} f)(s) = \sum_{i=0}^{k-1} f(i) \cdot \mathbf{x}_{s-d \cdot i}, \quad (4)$$

where $*$ is the convolution operator, d is the dilation factor, k is the filter size, and $s - d \cdot i$ denotes the past input. A TCN was applied in [39] to capture temporal dependencies among cellular traffic data.

In addition, residual and skipped layers are incorporated to handle gradient disappearing issues. Consequently, the combination of a ResNet and dilated convolution of a TCN enables deeper network layers and more extensive historical data for capturing the intricate temporal connections in cellular traffic. However, CNN-based methods primarily rely on local receptive fields for feature extraction, which may lead to insufficient

capture of global information, especially in tasks involving comprehensive contextual understanding. These limitations can be alleviated by the attention-based models to be introduced subsequently.

Attention-based methods

In recent years, attention mechanisms have made considerable advances in a variety of deep learning task domains. The primary objective of self-attention is to determine the degree of association between each vector and other vectors. The standard self-attention formula is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (5)$$

where Q, K, V are the linear variations of the input $\mathbf{X}$ as follows:

$$\begin{aligned} Q &= \mathbf{XW}^Q, \\ K &= \mathbf{XW}^K, \\ V &= \mathbf{XW}^V, \end{aligned} \quad (6)$$

where $\mathbf{W}^Q \in \mathbb{R}^{d_{model} \times d_q}$, $\mathbf{W}^K \in \mathbb{R}^{d_{model} \times d_k}$, and $\mathbf{W}^V \in \mathbb{R}^{d_{model} \times d_v}$ are trainable parameters. Through training, the linear transformation of $\mathbf{X}$ can be projected into various representation subspaces to increase the fitting ability of the model. The inner product of vectors describes the angle between 2 vectors and the projection of one vector onto the other. In (5), $\mathbf{QK}^T$ is a square matrix in which each row vector represents the result of the inner product of each vector with the other vectors. Let us assume $\mathbf{A} = \mathbf{QK}^T$, then the distribution of $\text{softmax}(\mathbf{A})$ will be related to d_k . After dividing each element in $\text{softmax}(\mathbf{A})$ by $\sqrt{d_k}$, the variance returns to 1, and the gradient value can remain stable throughout the training procedure.

Attention-based models have recently gained popularity in cellular traffic forecasting. Shen et al. [55] suggested an attention-assisted deep learning method for modeling complex patterns in extremely dynamic traffic flows. The self-attention layer is used in a time-embedding block that concatenates both the present and previous states, thus capturing long-term dependencies. Then, to investigate more useful characteristics from the self-attention layer, a feedforward neural network consisting of convolution operation, batch normalization, and rectified linear unit activation is introduced. A wireless network prediction method based on LSTM and attention structures is provided in [56]. Due to the coexistence of burst and periodicity in wireless cellular traffic data, it is frequently difficult to precisely estimate traffic sequences. For modeling sequence data, a feature extractor made up of LSTM units is used to extract complicated temporal properties. As long sequences may degrade the performance of RNN-based approaches, the intermediate output of the RNN is coupled to an attention network. For other related research adopting attention mechanisms, see [62].

In conclusion, RNN-based algorithms have demonstrated remarkable performance in sequential data prediction. However, they are insufficient for modeling extremely lengthy sequences and typically require a considerable amount of training time. CNN-based systems are easier and faster to train. In addition, attention-based approaches focus all past input states and treat them differently based on their attention scores, which increases the accuracy of prediction for long-term traffic data.

Spatial-temporal prediction

This section introduces deep-learning-based spatial-temporal prediction methods. The techniques for temporal dependency modeling discussed in the “Temporal prediction” section are also applicable to spatial-temporal prediction. Thus, we classify spatial-temporal cellular traffic prediction methods into 3 main categories: grid-, graph-, and attention-based methods, and we describe each accordingly. Grid-based methods are proposed for modeling Euclidean data, and graph-based methods are therefore used to model non-Euclidean spatial structure data. A comparison between Euclidean and non-Euclidean data structures is shown in Fig. 3. This section also provides a brief introduction to the GNNs implemented in spatial-temporal prediction to facilitate comprehension.

Grid-based methods

Grid-based methods are popular for spatial-temporal cellular traffic prediction, where the grid is generally a geographically defined grid. This strategy splits a metropolitan area into grids, each of which represents a region where traffic data are collected. The grid-based adjacency matrix usually depicts the adjacent relationship between regions and is based on the

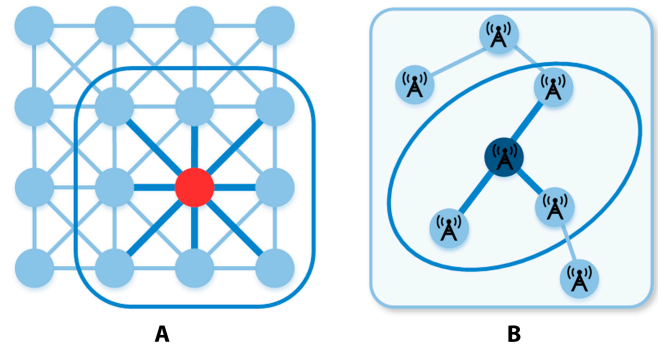


Fig. 3. Comparison between Euclidean and non-Euclidean data structures, reproduced with permission of Ref. [71]. Copyright of ©2023 IEEE. Reprinted, with permission, from *IEEE Transactions on Neural Networks and Learning Systems*. (A) Example of a grid-based structure where the features can be extracted with 2D convolution. The neighbors of each node are determined by the convolutional filter size, which are ordered and have a fixed size. (B) Example of a graph-based structure where the features can be extracted with graph convolution. In contrast to grid data, the neighbors of a graph node are unordered and variable in size.

premise that neighboring locations serve similar user groups and experience frequent user migration. The grid-based organization of traffic data resembles the structure of images [55,63]. Due to the intrinsic benefits of CNNs for exploiting image-like data, the majority of grid-based prediction methods are based on the CNN [21] or variations of the CNN, such as the 3-dimensional CNN (3D-CNN) [64] and ConvLSTM [20]. The temporal dependency modeling techniques for grid-based spatial-temporal forecasting are comparable to the temporal prediction methods described in the “Temporal prediction” section. Thus, we focus on grid-based spatial dependency modeling methodologies and the integration of temporal and spatial modeling methods for prediction in this subsection.

The CNN was initially proposed for the extraction of spatial features from image data, but has demonstrated astounding success in a wide variety of tasks [65]. Spatial-temporal traffic data can be viewed as video-like data, where each frame represents the traffic values of several nodes at a certain time step. Typically, CNN-based spatial-temporal prediction models comprise several components, including convolutional layers, a flattening layer, fully connected layers, and an output layer. The convolutional layer is used for the extraction and nonlinear transformation of spatial features. The flattening and fully connected layers are used to further transform the features, and the output layer is used to obtain the final prediction result. With a multi-layer structure, CNN-based models can capture the local and global spatial dependencies of traffic data hierarchically [38]. Zhang et al. [38] employed 2 parallel CNN-based networks to capture the temporal closeness dependence and period dependence of traffic data. Further, a learnable parametric matrix was implemented to fuse their generated features.

The 3D-CNN is an extension of the 2-dimensional CNN (2D-CNN) that adds the time dimension and works on 3-dimensional (3D) input. It is a common grid-based technique for predicting cellular traffic. 3D-CNNs can concurrently capture temporal and spatial dependencies, and the computational cost of training 3D-CNNs is lower than that of RNNs [64]. Typical 3D-CNN-based spatial-temporal prediction networks consist of 3D convolutional layers, dense layers, and an output layer, similar to 2D-CNN-based networks. A 3D-CNN maintains a 3D convolutional kernel that can directly extract local spatial-temporal features. Bega et al. [64]

proposed an encoder–decoder-based model called DeepCog, which stacks multiple 3D convolutional layers as the encoder. The decoder, which comprises several fully connected layers, leverages the pertinent features extracted by the encoder to produce the final forecast results.

CNNs can also be combined with RNNs for spatial–temporal prediction. This combination benefits from the advantages of CNNs for spatial dependency modeling and RNNs for sequence modeling. Here, we focus on a classical CNN–RNN structure, ConvLSTM, which is a combination of CNN and LSTM. ConvLSTM was first proposed for precipitation forecasting [20] and extends the fully connected LSTM (FC-LSTM) to handle spatial–temporal data. ConvLSTM has also been widely used for mobile traffic prediction and has shown promising prediction performance [3,19,66,79]. ConvLSTM extends the input and state vectors of FC-LSTM from 2D matrices to 3D tensors. Specifically, ConvLSTM is designed for 3D input data by replacing the full connection structures in both the state-to-state and input-to-state transitions of FC-LSTM with convolutional structures. Therefore, the future cell state of ConvLSTM is jointly determined by the current and past states of the grid itself and its spatial neighbors. The key formula of ConvLSTM is as follows:

$$\begin{aligned} f_t &= \sigma(\mathbf{W}_{xf} * \mathcal{X}_t + \mathbf{W}_{hf} * \mathcal{H}_{t-1} + \mathbf{W}_{cf} \odot C_{t-1} + \mathbf{b}_f), \\ i_t &= \sigma(\mathbf{W}_{xi} * \mathcal{X}_t + \mathbf{W}_{hi} * \mathcal{H}_{t-1} + \mathbf{W}_{ci} \odot C_{t-1} + \mathbf{b}_i), \\ \tilde{C}_t &= \tanh(\mathbf{W}_{xc} * \mathcal{X}_t + \mathbf{W}_{hc} * \mathcal{H}_{t-1} + \mathbf{b}_c), \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t, \\ o_t &= \sigma(\mathbf{W}_{xo} * \mathcal{X}_t + \mathbf{W}_{ho} * \mathcal{H}_{t-1} + \mathbf{W}_{co} \odot C_t + \mathbf{b}_o), \\ \mathcal{H}_t &= o_t \odot \tanh(C_t), \end{aligned} \quad (7)$$

where $*$ denotes the convolution operator, and $\mathcal{X}_t$ represents the input 3D data. C_{t-1} and $\mathcal{H}_{t-1}$ represent the cell state and hidden state at time step $t-1$, respectively, and are also 3D tensors. $\mathbf{W}_*$ and $\mathbf{b}_*$ represent the weight and bias parameters, respectively. f_t , i_t , and o_t denote the forget, input, and output gates, respectively, which maintain the same function as in LSTM but in 3 dimensions.

Zhang et al. [19] incorporated ConvLSTM as the primary component for capturing the complex patterns in cellular traffic data. Evaluation experiments on real datasets demonstrated the superior performance of the proposed model in cellular traffic prediction. Zhang and Patras [66] implemented both 3D-CNN and ConvLSTM to encode the spatial–temporal features of cellular traffic. Then, the outputs of the 2 components were fused and decoded into long-term traffic predictions. Wu et al. [3] applied 2 ConvLSTMs to model the geographical and semantic spatial–temporal correlations of cellular traffic and combined them for accurate traffic demand prediction, which enabled data-driven base station sleep control in fifth-generation mobile networks (5G). Some approaches also integrate external factor modeling for grid-based cellular traffic prediction [19,55,67,79]. We believe that external factors have an important impact on traffic prediction, as will be discussed in the “Future directions” section.

Grid-based approaches have played a vital role in enhancing the modeling of spatial–temporal dependencies and have shown remarkable performance in traffic prediction. However, they also have limitations in making precise network element-level predictions due to the coarse granularity of the typical

grid topology. Grid-based approaches build spatial relationships mostly on the basis of spatial proximity. This type of spatial structure makes it difficult to capture spatial dependencies over long distances. In contrast, graphs can provide a more realistic representation of the spatial dependencies in real-world cellular traffic data. This scenario lends itself to graph-based approaches, which are discussed in the following subsection.

Graph-based methods

Previous methods, including CNNs and ConvLSTM, are restricted to Euclidean data. Graphs are a type of data structure where each node has a specific number of unordered neighbors. GNNs are deep-learning-based algorithms that operate on graph structures and are regarded as state-of-the-art techniques for spatial–temporal cellular traffic prediction. Inspired by the success of CNNs in multi-domain tasks, graph convolution networks (GCNs) specify local convolution filters on the graph structure and implement the transformation of CNNs from Euclidean to non-Euclidean domains. This subsection first covers the basics of GCNs and then describes how GNNs are used to predict cellular traffic. GCNs are the most widely used and have the largest number of variants for cellular traffic forecasting. A basic schematic of a GCN is shown in Fig. 4. GCNs are generally classified into spectral- and spatial-based approaches.

Spectral graph convolution: The spectral graph convolution process is defined by computing the eigendecomposition of the graph Laplacian in the Fourier domain. In the field of spectral graph theory, a graph is commonly depicted as a combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{A}$ is the adjacency matrix of the graph, and $\mathbf{D}$ is the degree matrix of vertices, $\mathbf{D}_{ii} = \sum_j (\mathbf{A}_{ij})$. The normalized form of $\mathbf{L}$ is defined as $\mathbf{L}^{sym} = \mathbf{D}^{-\frac{1}{2}} \mathbf{L} \mathbf{D}^{-\frac{1}{2}} = \mathbf{I}_N - \mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}}$. $\mathbf{L}^{sym}$ is a positive semidefinite matrix that can be factorized as $\mathbf{L}^{sym} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$, where $\mathbf{U}$ is the Fourier basis whose columns are corresponding eigenvectors, and $\mathbf{\Lambda}$ is a diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{L}^{sym}$. Bruna et al. [69] first defined the convolution operation as the multiplication of a signal x with a filter g_θ using the convolution theorem and proposed the spectral network:

$$g_\theta *_{G^x} = \mathbf{U} g_\theta(\mathbf{\Lambda}) \mathbf{U}^T x. \quad (8)$$

The spectral network suffers from high computational complexity and non-spatially localized filters. Therefore, the Chebyshev polynomials $T_k(x)$ were adopted to approximately fit the convolution kernel in [70], where $T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x)$ with $T_0 = 1$, $T_1 = x$. The convolution operation can be rewritten as

$$g_\theta *_{G^x} = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathbf{L}}) x. \quad (9)$$

The propagation rule presented by Kipf and Welling [23] is simple and well-behaved, which limits the convolution operation to $K = 1$ and approximates $\lambda_{\max} \approx 2$. The graph convolution is defined as

$$g_\theta *_{G^x} \approx \theta \left(\mathbf{I}_N + \mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}} \right) x. \quad (10)$$

The spectral graph convolution transforms data from the spatial domain to the spectral domain for processing in accordance with the convolution theorem and graph signal processing (GSP). This technique can only be employed on undirected graphs, and the structure of the graph cannot be modified during training.

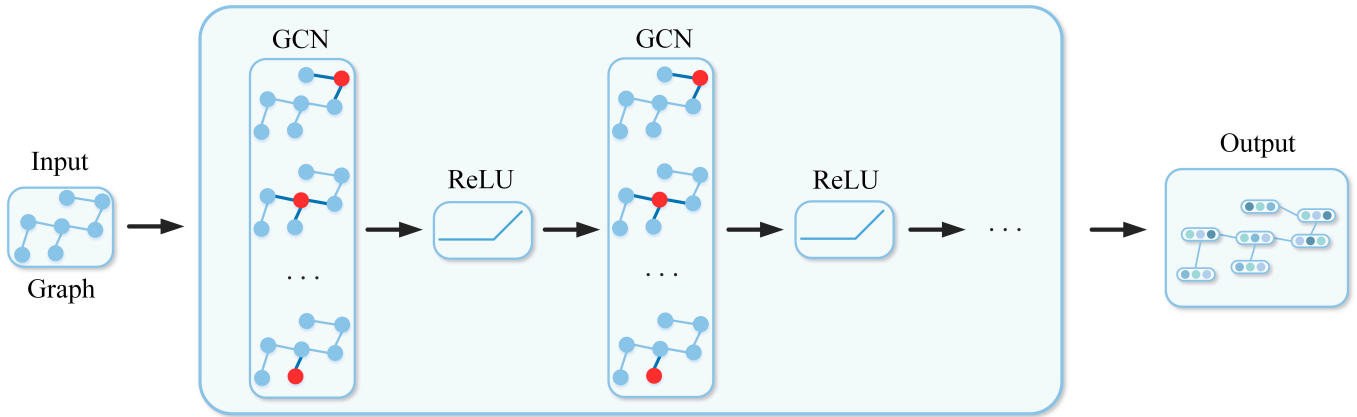


Fig. 4. Basic schematic of a GCN, reproduced with permission of Ref. [71], Copyright of ©2023 IEEE. Reprinted, with permission, from *IEEE Transactions on Neural Networks and Learning Systems*.

Spatial graph convolution: Spatial approaches define convolutions directly on neighbors that are spatially adjacent. The convolution operation on the graph entails 2 steps: establishing the neighborhood and evaluating the inner product of the nodes in the neighborhood with the convolution kernel parameters. See Fig. 4 for an illustration. Li et al. [72] treated graph convolution as a diffusion process and proposed the diffusion graph convolution network. They posited that information is transmitted from a node to one of its neighbors with a specific probability so that the information distribution is in equilibrium after a few circles. The authors defined the graph convolution as

$$\mathbf{H} = \sum_{k=0}^K f(\mathbf{P}^k \mathbf{X} \mathbf{W}^{(k)}), \quad (11)$$

where f is an activation function, $\mathbf{W} \in \mathbb{R}^{N \times F}$ is a weighted matrix, and $\mathbf{P} = \mathbf{D}^{-1} \mathbf{A} \in \mathbb{R}^{N \times N}$ is the transition matrix of the diffusion process. The neighborhood of a certain node at the k th step can be built based on $\mathbf{P}^{(k)}$. In [73], the authors proposed the graph attention network (GAT). The attention mechanism assigns a unique value to each nearby node to determine its importance to the center node. The GAT is built by stacking multiple graph attention layers, where the output features for each node can be calculated by

$$\mathbf{h}'_i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W} \mathbf{h}_j \right). \quad (12)$$

Here, α_{ij} indicates the importance of node n_j to node n_i , $\mathcal{N}_i$ is the first-order neighborhood of node n_i , $\mathbf{h}_i \in \mathbb{R}^F$ is the input feature of n_i , and $\mathbf{W} \in \mathbb{R}^{F \times F}$ denotes the linear weights matrix. The GAT is adapted for inductive learning and can process directed graphs.

Spatial graph convolution explicates the convolution operation on spatial neighbors and is adaptable and does not require a fixed graph structure. However, compared to spectral-based approaches, it raises potential concerns of over-smoothing [74].

In the field of cellular flow prediction, GCNs have been used to model the complicated spatial dependencies of cells and are frequently combined with other approaches that model temporal dependencies to increase prediction accuracy. A common combination is RNN-based methods and GCN, where matrix multiplication in RNN-based methods is replaced with graph convolution. Fang et al. [76] investigated a graph convolutional LSTM model that combines the GCN and LSTM for both spatial and temporal information. Experiment results suggest that this

spatial-temporal prediction strategy is superior to either GCN or LSTM alone and more dependable for medium- and long-term prediction. Lin et al. [4] combined spatial-temporal prediction techniques, including a multi-graph convolutional network for spatial features and a multi-channel LSTM for temporal patterns. Based on the predicted traffic data and a learned capacity model, an optimal base station sleeping strategy was proposed to minimize network power consumption. Yu et al. [29] presented STEP, a “spatio-temporal fine-grained user traffic prediction mechanism”. It employed a novel deep graph convolution network that combines GCN with GRU to efficiently learn from massive user datasets. In addition to RNN-based methods, convolutional structures can also be combined with a GCN for spatial-temporal correlation modeling. The convolution-based structure solves the time consumption problem of RNN-based methods due to the complex gating mechanism and is comparable to or better than the classical recurrent network in sequence modeling tasks. Figure 5 depicts a fundamental spatial-temporal prediction architecture in which a GCN models spatial dependencies and a 1D CNN models temporal dependencies. The GCN and 1D CNN can be replaced with alternative spatial-temporal modules for various prediction tasks. Piovesan et al. [26] utilized a 1D CNN to obtain high-level temporal characteristics and graph convolutions to get spatial features from nearby nodes. Zhao et al. [27] designed a stacked residual structure integrating graph convolutions and gated linear units (GLUs) to capture the spatial-temporal features of mobile traffic. In the GLU, the input traffic sequence is first fed into a convolutional layer, which is followed by a sigmoid gating mechanism that controls which information can be sent to the subsequent layer. For further related research, refer to [39].

Because information propagation occurs simultaneously along the spatial and temporal dimensions, there is a third relationship in the spatial-temporal problem, called the synchronous spatial-temporal correlation. Wang et al. [28] constructed a spatial-temporal joint graph convolution kernel, which holds a spatial-temporal receptive field for synchronous modeling. Zhao et al. [92] devised an aggregation spatial-temporal adjacency matrix that takes into account both the spatial relationship between the nodes of the current timeslot and the temporal correlation across T consecutive historical time steps. Then, an aggregation graph convolution is adapted to model spatial-temporal dependencies synchronously. However, the investigation

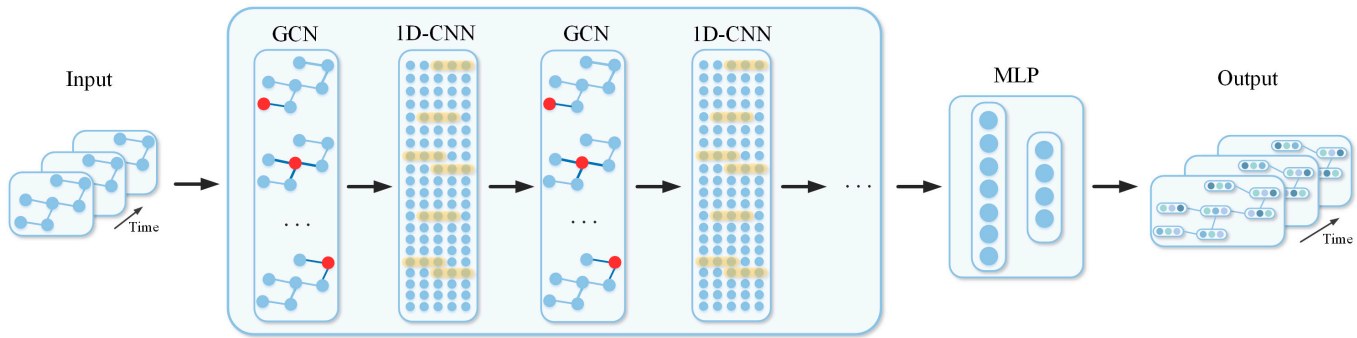


Fig. 5. Typical GNN structure (GCN+1D CNN) for spatial-temporal modeling, where the GCN is used to capture spatial dependencies between nodes, and the 1D CNN is used to capture the temporal dependencies of traffic data.

of synchronous spatial-temporal correlation modeling is in its early stages.

In general, graph-based approaches allow for the depiction of fine-grained spatial relations between cells and portray mobile data as a graph structure. In addition, graph-based methods overcome the limitations of grid-based methods, which can only construct spatial correlations based on distance, and can learn graph structures via node similarity, traffic handover, and adaptively through training. Graph-based approaches also have disadvantages, such as the fact that graph construction has a considerable impact on prediction accuracy and training graph-based models is sometimes time-consuming.

Attention-based methods

In recent years, attention mechanisms have been recommended by the vast majority of researchers because they are highly parallelizable and can enhance training effectiveness considerably. The most important benefit of the attention mechanism is its capacity to explore both global and local relationships. The attention mechanism assigns a distinct weight parameter to each object of the input, imparting greater weight to objects that are related to the input and suppressing irrelevant data. In the cellular traffic prediction problem, the attention mechanism can model the importance of neighboring nodes to the central node in the spatial dimension and the importance of different historical time slices in the temporal dimension.

To investigate the relationships between local and distant areas of a city, 2 attention methods for spatial dependency modeling were utilized in [77]. The areas are clustered into several classes, and an attention mechanism is utilized to represent the influence of the different neighbors within a class. Meanwhile, another attention mechanism is employed to combine the relevant neighbor information gathered previously from other classes with the temporal characteristics of the target node itself to predict mobile traffic. This strategy integrates different effects holistically. Further, attention mechanisms can be used to model spatial-temporal dependencies. In [78], a spatial attention module was employed to gather correlations between various regions, and a temporal attention module was designed to acquire correlations between various slices of historical mobile traffic data. The prediction performance demonstrates that the proposed model can capture dynamic spatial-temporal features effectively.

The incorporation of the attention mechanism, which explores both global and local relationships, considerably enhances the accuracy of predictions. In addition, compared to classical recurrent networks, the computation of the attention mechanism

does not depend on the result of the previous steps. Thus, it has gained widespread acceptance in the field of cellular traffic forecasting. For more related studies, see [79,80].

This subsection presents a discussion on 3 categories of spatial-temporal prediction methods: grid-, graph-, and attention-based approaches. Grid-based prediction methods, predominantly implemented using CNNs, are widely popular. They have shown superior performance compared to sequence modeling methods that overlook spatial dependencies. However, graph-based methods generally outperform grid-based methods as they can effectively capture spatial relationships among non-Euclidean data structures. These methods offer more flexibility by not being restricted to the strong adjacent neighbor connections in a grid. Attention-based approaches can capture global spatial and temporal dependencies, resulting in improved prediction outcomes. In addition, for multivariate prediction, most deep-learning-based methods adopt the approach of incorporating multiple variables as input features into deep neural networks [6,28]. Through network training, these methods implicitly model the interdependencies among the variables. This review serves as a summary of current methodologies. As the field continues to evolve rapidly, ongoing research is expected to bring additional insights and advancements.

State-of-the-art performance

Before discussing the model performance comparison, the generic evaluation metrics of traffic prediction are introduced and summarized. The performance of a prediction model is usually evaluated based on the error between the ground truth and predicted values, as well as the determination coefficient. Mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and the determination coefficient are frequently cited in the literature on traffic prediction as evaluation metrics derived from observation. The formulas of these metrics are as follows:

$$\begin{aligned}
 \text{MAE}(\mathbf{y}, \hat{\mathbf{y}}) &= \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \\
 \text{RMSE}(\mathbf{y}, \hat{\mathbf{y}}) &= \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \\
 \text{MAPE}(\mathbf{y}, \hat{\mathbf{y}}) &= \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%, \\
 R^2 &= \frac{\sum_{i=1}^N (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^N (y_i - \bar{y})^2},
 \end{aligned} \tag{13}$$

where y_i is the ground truth, $\hat{y}_i$ is the predicted value, N is the number of observations, and $\bar{y}$ is the mean of the ground truth. For the evaluation metrics MAE, RMSE, and MAPE, the smaller the value, the greater the prediction effect. The determination coefficient, which is equal to the square of the correlation coefficient, has a range of 0 to 1. The greater its value, the better the effect.

In this section, we summarize the state-of-the-art prediction performance from the surveyed studies, which used publicly available datasets, aiming to provide a baseline for follow-up works for model performance evaluation. Because the Telecom Italia dataset (which includes data from Milan and Trentino) is intensively used in cellular traffic prediction, the state-of-the-art performance on this dataset is the focus of the comparison presented in Table 2. Readers interested in performance on other open datasets can check the related studies mentioned in the “Datasets” section for details. Because the differences in data preprocessing and experimental settings have an important impact on model evaluation, evaluation results with denormalized data are chosen for comparison from the Milan dataset to eliminate such biases. While there are fewer studies using the Trentino dataset for model evaluation, the most commonly used normalized data are selected here to compare model performance. Further, different grid sizes, time intervals, and prediction windows also affect the evaluation of traffic prediction results. Therefore, we provide the state-of-the-art performance on 3 data subsets: SMS, call, and internet, with different experimental configurations. As shown in Table 2, for the same data subset, the corresponding experimental configurations and evaluation results are listed in ascending order of grid size. Note that we have specified a grid size of 100×100 for studies that did not state a grid size.

According to Table 2, we can make the following observations: First, the evaluation results on the same subset with different grid sizes are different. Second, for a given time interval on the same data subset, the performance decreases as the prediction window expands. For example, in view of the evaluation results of the call data subset with grid size 20×20 , the MAE with a prediction window of 60 min is lower than that

with a prediction window of 180 min. Moreover, for the SMS data subset with 10-min time granularity, the RMSE of multi-step prediction (60 min) is larger than that of single-step prediction (10 min). Third, for the same data subset, as the time granularity becomes coarser, the values of the evaluation metrics MAE and RMSE get larger. We believe that this is mainly because the data value itself increases after aggregating 10-min data to 1-h data. Through data aggregation on the time dimension, cellular traffic data tend to become relatively smooth with fewer fluctuations. From this perspective, the prediction accuracy of traffic data with coarse time granularity is generally better than that of traffic data with fine time granularity.

Note that as some of the literature fails to clearly state the experimental configurations, readers are requested to check the prediction results critically. With this in mind, it is necessary to build benchmarks for cellular traffic prediction. This would enable fair comparison of different models by unifying the experimental configurations on the specified datasets. We further explore future directions with respect to benchmarks in the “Future directions” section.

Open-source codes

In this part, we summarize the available open-source cellular traffic prediction implementations and provide links to their codes in Table 3. With the rapid growth of the cellular prediction field, an increasing number of models have been developed; however, only a small fraction of the literature has disclosed their model codes. Due to the fact that different studies use distinct datasets, experimental settings, and training approaches that may not be stated explicitly, it is difficult to replicate their results. According to Table 3, this issue has started to abate in recent years, with 3 studies in 2021 open-sourcing their codes. The PyTorch framework is widely used in deep learning methods of cellular traffic prediction.

Applications

In this section, we focus on the applications of cellular traffic prediction. The results of cellular traffic prediction derived

Table 2. State-of-the-art results for Telecom Italia [35] with different spatial grid sizes and time intervals

Open dataset		Grid size	Time interval	Prediction window	MAE	RMSE
Milan	SMS	20×20	60 min	60 min	15.165 [92]	43.015 [92]
		30×30	10 min	60 min	29.84 [28]	57.73 [28]
		100×100	10 min	10 min	-	24.2692 [59]
		100×100	60 min	60 min	21.8829 [79]	38.3087 [79]
	Call	20×20	60 min	60 min/180 min	8.61 [92]/9.8566 [81]	20.47 [92]/-
		100×100	10 min	10 min	-	13.72403 [59]
		100×100	60 min	60 min	13.1964 [79]	20.9656 [79]
	Internet	15×15	10 min	60 min	21.57 [82]	45.77 [82]
		20×20	60 min	60 min	85.99 [92]	172.03 [92]
		100×100	60 min	60 min	84.8507 [79]	129.1509 [79]
Trentino	SMS	10×10	60 min	60 min	0.6920 [83]	-
	Call	10×10	60 min	60 min	0.4281 [83]	-
	Internet	10×10	60 min	60 min	0.7471 [83]	-

Table 3. Open-source codes for cellular traffic prediction

Publication	Year	Framework	Link
Zhang et al. [83]	2021	PyTorch	https://github.com/chuanting/FedDA
Yao et al. [84]	2021	PyTorch	https://github.com/qlab2019/MVSTGN
Liu et al. [81]	2021	PyTorch	https://github.com/liuqingyao11/ST-Tran
Zhang et al. [19]	2019	PyTorch	https://github.com/chuanting/STCNet
Zhang et al. [38]	2018	PyTorch	https://github.com/chuanting/STDenseNet
Huang et al. [82]	2017	TensorFlow	https://github.com/IPCLab/CDRF

from various spatial-temporal modeling techniques could enhance intelligent 5G network construction and resource management, thus enhancing the quality of experience (QoE) for users. We divide cellular traffic prediction applications into 3 categories: routing, scheduling, and resource allocation.

Routing

Using a neural network framework, Isaacman et al. [85] explored route optimization and settings in the context of knowledge-defined networks (KDNs). To improve quality of service (QoS), they employed an offline reinforcement learning (RL) technique called deep deterministic policy gradient (DDPG). Given the traffic requests of flows as the state and the weights of links between traffic flows as the action, the reward functions are defined as QoS statistics, such as packet loss and latency. The fully connected layer is then replaced by a convolutional layer to extract the interrelationships between different traffic flows. Simulations in [85] on a network topology with 24 nodes indicate that the replacement could adequately capture the mutual effects of data flows and substantially reduce network latency.

Scheduling

Chinchali et al. [86] examined scheduling cellular network traffic flow generated by internet of things devices. An adaptive traffic optimizer was trained using DDPG, an RL algorithm that can handle a continuous action space. The agent receives the predicted network throughput as the reward function, which is implemented using a random forest model, together with the predicted network congestion metric, session numbers, cell efficiency, and some temporal data as the network state. Australian city-wide network data for a 4-week period was used to evaluate the DDPG model. Because this was the first time that data were collected without any benchmark in place, a stochastic model predictive control (MPC) was carefully created for comparative purposes. The utilization gain in mobile networks might be greatly increased by the RL-based scheduler, according to experimental results showing an increase of 14.7%. As a result, the RL scheduler produces more than twice as much traffic as the heuristic model.

Resource allocation

In recent decades, resource allocation and management have become more popular. The study in [87] examined how to estimate video traffic to manage resource allocation and reduce system cost. The authors created a discrete wavelet decomposed neural network architecture with the goal of reducing frequency components and traffic burstiness. A backpressure technique

was presented to simplify resource allocation in space-based information networks based on the anticipated traffic. The outcomes of the simulations demonstrate that the suggested approach can achieve excellent prediction accuracy while also considerably lowering system latency and power consumption.

Challenges and Future Directions

Finally, we explore the challenges and potential research areas of cellular traffic prediction in detail and provide a roadmap for future researchers. First, we describe some of the existing and unresolved problems and obstacles, including data quality and privacy, spatial-temporal modeling, and heterogeneity. Then, we discuss a number of future research directions—associated with external factor modeling, transfer learning, federated learning, and model interpretability—that are promising for solving the main challenges of cellular traffic prediction.

Challenges

Data quality and privacy

Data quality concerns are a major challenge in the field of cellular traffic prediction. Most research works directly with processed data, with little emphasis on addressing data quality issues. Problems with cellular traffic data include missing, noisy, and anomalous data. These issues can be influenced by multiple factors, such as defects in the process of acquiring equipment, base station energy conservation, and base station retirement. These problems have not been adequately addressed. Cellular data collected from individual mobile devices like smartphones provides a more detailed view of cellular network performance and usage at an individual user level [88]. These data are important for predicting cellular traffic, optimizing network performance, improving QoS, and enabling personalized services. However, collecting and processing this sporadic data can be challenging as it can be difficult to identify user-specific traffic patterns. In addition, selecting subscribers as examples of typical behavior is still an unresolved issue. Furthermore, these data comprise sensitive information on a number of facets of subscribers' lives, including their residences, workplaces, and areas of interest. Therefore, protecting users' privacy is an important concern.

Spatial-temporal modeling

As stated previously, cellular traffic data involve intricate temporal and spatial dependencies. Despite the fact that certain difficulties have been resolved by relevant efforts ([19,76]), there are still several obstacles in this regard. Typically, cellular traffic data have a broad size range. The peak traffic data of a

base station can surpass 1 GB, which is over a million times greater than its trough traffic. Although previous deep learning algorithms, such as CNN and LSTM, have achieved acceptable performance, their precision is insufficient to serve a network scheduling application. Modeling time-series data with such substantial changes still presents some obstacles. Although the commonly used GCN approach can introduce the influence of surrounding neighbors from the spatial dimension, it often relies on the development of a graph adjacency matrix, and differing graph construction methods have a considerable impact on the precision of spatial-temporal modeling. For instance, it is challenging to represent the dependence on distance space for a graph based on distance. Convenient traffic enables users to roam throughout a substantial portion of the city, and cellular traffic frequently demonstrates the spatial correlation of the entire network. Multi-layer GNNs can broaden the spatial receptive field and attempt to mimic the whole, but this typically causes the problem of oversmoothing. Because cellular traffic is influenced by the individual and collective actions of users, modeling spatial-temporal correlation in traffic data remains a formidable obstacle.

Heterogeneity

Due to the differences in geographic locations, user groups, surrounding environments (e.g., points of interest [POIs] and business areas), and network devices (e.g., wide-area and local-area stations) of wireless base stations, generated traffic frequently exhibits diverse patterns (heterogeneity) [28,83]. The intuitive solution to heterogeneity is to train an individual model for the traffic pattern of each node. This strategy would incorporate a large number of parameters, making it susceptible to overfitting and difficult to implement in practical applications [28]. Ignoring heterogeneity may severely degrade the performance of prediction methods [19]. In addition to spatial-temporal traffic data, traffic prediction also entails external factors such as weather, accidents, and social activities. Consequently, heterogeneous data integration is a typical challenge for traffic forecasting. Although some existing prediction methods based on transfer learning [19], federated learning [83], and model adaptation [89] have been proposed for modeling heterogeneity, the problem is still a long way from being solved. Typically, heterogeneity and the problem of large-scale network elements coexist. Consequently, cellular traffic prediction for large-scale and heterogeneous cellular networks is a considerable challenge for existing technologies.

Long-term traffic prediction

Typically, network planning entails the forecasting of prolonged future traffic sequences. Currently, the majority of deep-learning-based algorithms are appropriate for short- and medium-term forecasting. In addition, sequence-modeling-based methods for long-term traffic forecasting are frequently accompanied by error accumulation. Several works, including the transformer-based informer [90], have been proposed for forecasting generic long-term time series. However, the severe fluctuations in the long-term range of traffic data mean that long-term traffic prediction in cellular networks remains challenging.

Future directions

Benchmarking cellular traffic prediction

There are relatively few open-source codes in the field of cellular traffic prediction. As described in the “Open-source codes”

section, only 6 publications provide open-source codes that can be used to reproduce the performance of the proposed methods. Given that the majority of articles employ drastically different experimental settings, such as univariate or multivariate prediction, open-source code is necessary for the advancement of research in this field. It enables researchers to validate newly proposed algorithms more efficiently and precisely. As the field has evolved, more deep-learning-based approaches have been proposed, but there is no reproducible benchmarking framework for cellular traffic prediction. Even if articles utilize the same dataset, the multiple experimental settings make it difficult to compare the performance of different models. An important direction for future work is to conduct a fair and systematic comparison of multiple models using standard datasets under consistent experimental settings. A benchmark can provide future researchers with a reproducible baseline framework for an effective and objective evaluation of proposed methods, thereby preventing the duplication of efforts.

External factor modeling

Cellular traffic is impacted not only by spatial and temporal characteristics but also by multiple external factors. They are widely classified as time attribute and cross-domain factors. Time attributes, such as day of week and hour of day, are frequently utilized and closely related to cellular traffic data. In addition, holiday information, such as information about major celebrations, could benefit traffic forecasting, as the spatial-temporal traffic patterns on holidays are typically distinct from those on non-holidays. Meanwhile, cross-domain characteristics, such as POIs, base stations, weather, and social activities, also facilitate accurate prediction. These variables may aid in the modeling of various traffic usage patterns and have an impact on traffic utilization from distinct viewpoints.

Time attribute factors. Time attribute factors refer to the intrinsic time characteristic of each timestamp. Zhao et al. [91] modeled weekday and holiday effects on cellular traffic data as binary information. A 2-layer neural network was then designed to concatenate the external factors and input data to further examine the internal traffic patterns. Hour of day and holiday information were used by Zhang et al. [92] to better capture the temporal dependencies in cellular traffic. The time features were represented as vectors of 24 and 1 dimension, for the hour of day and the holiday, respectively. The effectiveness of the time embedding module in assisting traffic forecast has been further proved by simulation data.

Cross-domain factors. In spatial-temporal traffic forecasting, cross-domain factors are frequently used to examine spatial/regional relationships. Kuber et al. [93] extracted crucial characteristics from non-cellular auxiliary data, including POIs and precipitation data. Using an open-source application programming interface, the original cellular traffic was enhanced with features from surrounding cells and external sources. In a series of comparative experiments, the efficacy and superiority of this enhancement were further validated. Zhang et al. [19] analyzed 3 cross-domain datasets concerning base stations, POIs, and social activities that have been demonstrated to be strongly linked with cellular traffic. To delve more thoroughly into the important patterns of external data, they proposed a 2-layer CNN to extract features from the concatenated cross-domain data. To improve the prediction accuracy, Shen et al. [55] added cross-domain data (base stations, POIs, and social behaviors) to the suggested network structure, the time-wise attention-assisted

CNN. To extract features from external data affecting traffic changes, convolution layers were used. Additionally, date information was combined with the attention mechanism to investigate long-range temporal dependencies.

Transfer learning and federated learning

An unresolved issue in the field of cellular traffic analysis is model generalization. Transfer learning is a feasible approach to solving this problem. This method allows the model to be trained on a task and then transferred to other comparable tasks, thus eliminating the need for the model to be trained from scratch with a specific prior basis. The parameters produced from a fine-tuned model on the source domain can be shared as initial parameters with new models on the target domain to speed up the training process and reduce model complexity. Additionally, this method helps alleviate the data deficiency issue. In mobile network traffic analysis, when newly constructed stations lack sufficient historical traffic, transfer learning is used to train models by leveraging the experience of data-rich stations. Although there have been some efforts in the topic of cellular traffic prediction, pertinent research is still limited. Future possible transfer learning approaches should use a wider range of cross-domain data to improve the generalizability and precision of the models.

Federated learning is an emerging decentralized architecture for machine learning. It aims to achieve common modeling and enhance the performance of the central server while protecting the confidentiality of data for various local models. Federated learning permits participants to jointly model without sharing data, which can theoretically address the issue of data islands and limit the risk of data leakage to accomplish effective collaboration. The tremendous volume of traffic generated by billions of mobile devices necessitates modeling strategies that include both model efficiency and energy consumption. Due to the distributed learning paradigm, only gradient information is communicated between the local learners and the central aggregators in the context of cellular traffic prediction, which alleviates the capacity limitation and substantial communication overhead in wireless environments. Existing methods for federated learning either average the weights of local models or distinguish their contributions to the central model. Zhang et al. [94] suggested an enhanced federated learning strategy for wireless traffic prediction with weighted model aggregation embedded in spatial-temporal modeling. The proposed method outperformed other benchmarks when handling heterogeneous wireless traffic data. Federated learning on cellular traffic data is still in its infancy and should receive more academic attention in the future.

Model interpretability

Despite their exceptional performance in traffic analysis and prediction, deep neural learning techniques are frequently viewed as black boxes in terms of model interpretability. The Bayesian network, commonly known as the Bayesian net or belief network, was initially presented in [95] as a generative model within a probabilistic framework with inherent interpretability. It is a directed acyclic graph whose edges reflect the dependencies between 2 nodes (variables) and the directions describe their interactions. In the domain of transportation networks, the Bayesian network has demonstrated its effectiveness in traffic flow forecasting, where road segments are denoted as input nodes and the Pearson correlation coefficient is used to

screen out the most pertinent variable for prediction [96]. Despite the paucity of the literature on Bayesian networks for cellular traffic prediction, it is highly likely that their excellent interpretability will aid in the prediction of cellular traffic data and pave the way for practical implementations.

Conclusions

In this paper, we presented a comprehensive survey of cellular traffic prediction based on deep learning. Specifically, a classification of cellular traffic prediction problems and an overview of open traffic datasets were provided. Then, prediction methods based on deep learning for modeling the most essential temporal and spatial dependencies in cellular traffic were discussed in detail. We also comprehensively summarized the state-of-the-art performance on 2 popular open datasets with multiple data settings. In addition, the open-source codes in this domain were listed to facilitate future research. We also outlined the applications of cellular traffic prediction. Finally, we presented a brief overview of challenges and prospective future directions to help future researchers conduct further research. This survey provides a comprehensive review and in-depth analysis of the problems and solutions associated with deep-learning-based cellular traffic prediction. This paper aims to help readers in gaining a thorough comprehension of deep-learning-based traffic prediction algorithms and in selecting interesting and challenging research topics to pursue in this vital field.

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